

Chapter 21

Simulation and the Central Limit Theorem

Lecture 21 | Devore 9e: 5.4–5.5

Last time: samples, statistics, and the closed-form derivation of the sampling distribution of $\bar{X}$ for two exponentials. That derivation was already unpleasant. Today: what to do when it is impossible.

21.1 Numerical simulation

Closed form works when both the underlying distribution and the statistic are simple. It fails as soon as

- n is large — imagine the region $\{x_1 + \cdots + x_{100} \leq 100t\}$;
- the underlying distribution is complicated;
- the statistic is awkward, e.g. $|X_1| + |X_2| + \cdots + |X_n|$ or the rank of X_1 among $X_1, \dots, X_n$.

Simulation handles all of them. The setup:

- a statistic $T = T(X_1, \dots, X_n)$ — e.g. $\bar{X}$ or S ;
- a sample distribution, which we can *draw from* even if we cannot integrate against it (a black box);
- the goal: $F_T(t) = \mathbb{P}\{T \leq t\}$, or the pdf of T .

The algorithm. Repeat k times:

$$\begin{aligned} t_1 &= T(x_1, \dots, x_n) && \text{from a freshly generated sample} \\ t_2 &= T(x_1, \dots, x_n) && \text{from another fresh sample} \\ &\vdots \\ t_k &= T(x_1, \dots, x_n) \end{aligned}$$

Then histogram $t_1, \dots, t_k$. That histogram *is* an estimate of the pdf of T , and the proportion of t_i 's below t estimates $F_T(t)$.

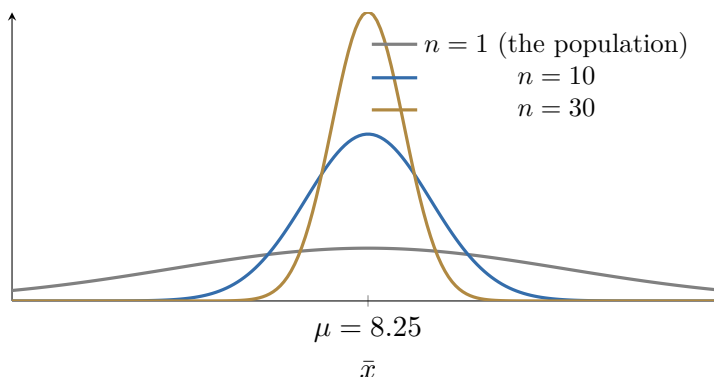
Example 21.1. Take $X_i \sim \text{Normal}(\mu, \sigma^2)$ with $\mu = 8.25$, $\sigma = 0.75$, sample size $n = 10$, and $k = 500$ repetitions. Each repetition gives one number $\bar{x}_j = (x_1 + \dots + x_{10})/10$; we get 500 of them — 8.0, 8.1, 7.9, ..., 8.6, 7.5 — and histogram them.

Remark 21.2 (Two knobs, two different effects).

larger k more *accuracy*: the histogram converges to the true pdf of T , whatever it is.

larger n the pdf of T *itself changes*: the peak stays at μ but the spread shrinks.

Confusing these two is the classic error. Increasing k tells you more about the same object; increasing n changes the object.



21.2 Mean and variance of the sample mean

Before the theorem, two computations that need no assumptions beyond i.i.d.

Theorem 21.3. If $X_1, \dots, X_n$ are i.i.d. with mean μ and variance σ^2 , then

$$\mathbb{E}[\bar{X}] = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}, \quad \text{SD}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

Proof. By linearity of expectation (which needs no independence),

$$\mathbb{E}[\bar{X}] = \mathbb{E}\left[\frac{X_1 + \dots + X_n}{n}\right] = \frac{\mathbb{E}[X_1] + \dots + \mathbb{E}[X_n]}{n} = \frac{n\mu}{n} = \mu.$$

For the variance we *do* need independence, which makes variances add:

$$\text{Var}(\bar{X}) = \frac{1}{n^2} \text{Var}(X_1 + \dots + X_n) = \frac{\text{Var}(X_1) + \dots + \text{Var}(X_n)}{n^2} = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}. \quad \square$$

Remark 21.4 (The $\sqrt{n}$ law). The precision of an average improves like $\sqrt{n}$, not like n . To halve your error bar you must *quadruple* the sample size. This single fact governs the cost of every survey and every experiment.

21.3 The central limit theorem

Simulation shows something striking: whatever you start with — Weibull, uniform, something lumpy and skewed — the histogram of $\bar{X}$ comes out bell-shaped once n is large enough.

Theorem 21.5 (Central limit theorem). *Let $X_1, X_2, \dots, X_n$ be i.i.d. with mean μ and variance $\sigma^2 < \infty$. Then for large n ,*

$$\bar{X} \stackrel{\text{approx}}{\sim} \text{Normal}\left(\mu, \frac{\sigma^2}{n}\right), \quad \text{equivalently} \quad \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \stackrel{\text{approx}}{\sim} \text{Normal}(0, 1).$$

The mean and variance are exactly what we computed above; the content of the theorem is that the *shape* is normal, regardless of the underlying distribution.

Remark 21.6 (Rule of thumb). $n > 30$ is the usual working threshold. It is only a rule of thumb: a nearly symmetric underlying distribution needs far fewer, and a heavily skewed one (or one with a heavy tail) needs far more. Note the hypothesis $\sigma^2 < \infty$ — for the heavy-tailed pmf of Lecture 9, where even the mean is infinite, the theorem simply does not apply.

Example 21.7 (Why this matters). $X_1, \dots, X_{1000}$ are i.i.d. from some complicated Weibull. What is $\mathbb{P}\{\bar{X} \leq t\}$? Deriving it exactly is hopeless. By the CLT,

$$\mathbb{P}\{\bar{X} \leq t\} \approx \Phi\left(\frac{t - \mu}{\sigma/\sqrt{1000}}\right),$$

which is a table lookup.

Example 21.8 (Normal approximation to the binomial, explained). $X \sim \text{Binomial}(n, p)$ is the sum of n i.i.d. Bernoulli variables, each with mean p and variance $p(1 - p)$. Applying the CLT to that sum recovers exactly the approximation quoted in Lecture 14:

$$X \stackrel{\text{approx}}{\sim} \text{Normal}(np, np(1 - p)).$$

Example 21.9 (How much data does an edge need?). A coin lands heads with probability 0.51 instead of 0.5. How many flips do we need before the excess of heads stands out, say by two standard deviations?

The proportion of heads $\hat{p} = \bar{X}$ has standard deviation $\sqrt{p(1 - p)/n} \approx 0.5/\sqrt{n}$; the standard deviation of an average like this is often called its **standard error**. The edge is 0.01, and asking for $0.01 \geq 2 \times 0.5/\sqrt{n}$ gives $\sqrt{n} \geq 100$, so $n \geq 10,000$ flips. For scale: a fair coin flipped 10,000 times gives 5000 heads on average, with standard deviation $\sqrt{10,000 \cdot \frac{1}{4}} = 50$, so 5100 or more heads is two standard deviations high and has probability about $1 - \Phi(2) \approx 0.023$.

The same arithmetic applies to anything noisy. A trading strategy earns 0.05% a day on average, with a daily standard deviation of 1%. After a year of 252 trading days, the average daily return has standard error $1\%/\sqrt{252} \approx 0.063\%$, so the true mean sits only about 0.8 standard errors from zero: a year of data cannot tell this strategy apart from luck. Two standard errors needs $0.05 \geq 2 \times 1/\sqrt{n}$, that is, $n \geq 1600$ days, over six years.

21.4 The exactly-normal case

When the sample is drawn from a normal population, the approximations become identities and the sample variance gets a distribution too.

Theorem 21.10. *Let $X_1, \dots, X_n$ be i.i.d. $\text{Normal}(\mu, \sigma^2)$. Then*

1. $\bar{X} \sim \text{Normal}\left(\mu, \frac{\sigma^2}{n}\right)$ exactly, for every n ;
2. $\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$;
3. $\bar{X}$ and S^2 are independent.

Remark 21.11. Part (2) is where the $n - 1$ finally earns its keep, and where the name “degrees of freedom” comes from: the n deviations $X_i - \bar{X}$ satisfy one linear constraint, $\sum_i (X_i - \bar{X}) = 0$, so only $n - 1$ of them are free. Part (3) — that the sample mean and the sample variance are independent — is special to the normal distribution, and it is what makes the t -statistic work in a later course.